CSE 2215: Data Structure and Algorithms-I

1. Draw a max-heap and min-heap trees from the following data where ID=last two digits of your student ID.

   10 40 20 8 99 ID 15 17

2. Sort the data in Ques-1 using ascending order and descending order heapsort.

3. Discuss the time complexity of Max-Heapify(A, i, n) or Adjust (A, i. n) procedure in HeapSort(A, n) algorithm

4. Discuss the time complexity of HeapSort(A, n) algorithm

5. How can you find the indices of left and right child in array with respect to father in a max heap tree?

6. Find the time complexity of the following algorithms.

a)	sum=0;
	k=10;
	for(i=1; i<=n; ++i){
	  for(j=3; j<=n; ++j){
	    sum=sum+j;
	    k=k-1;
	  }
	  printf(“%d”, sum);
	}
	printf((“%d %d”, sum, k);
		b)	sum=0;
			k=0;
			while(k<=n){
			  sum=sum+k;
			  printf(“%d %d”, sum, k);
			  k=k+1;
			}
			printf(“%d”, sum);

c)	void sample(int n){
	  if (n>1){
	    sample(n-1);
	    printf(“%d”, n);
	    sample(n-1);
	  }
	}
		d)	void sample(int n){
			  if (n>1){
			    sample(n/2);
			    printf(“%d”, n);
			    sample(n/2);
			  }
			}

7. Prove that
   i) $f(n)= 2n+3= O(n)$
   ii) $f(n)= 3n^2-4n+5= \Theta(n^2)$
   iii) $f(n)= 2n-3=\Omega(n)$
   iv) $f(n)=2n-3=o(n^3)$

8. Show the mechanism of the following ascending/descending order sorting algorithms for the following data where ID is last one digit of your student ID.

10	40	20	8	99	ID	15	17

i) Bubble Sort
ii) Selection Sort
iii) Radix Sort
iv) Bucket Sort